A

Song sparrow
Swamp sparrow
Savannah sparrow
White−throated sparrow
Red−winged blackbird
Common yellowthroat
Chaffinch
House finch
House sparrow
Zebra finch
European robin
Swainson's thrush
White−throated dipper
Common myna
Eurasian blackcap
Garden warbler
Silvereye
Barn swallow
Black−capped chickadee
New Caledonian crow
Blue jay
Paradise−crow
Rifleman
Blue−and−yellow macaw
Swift parrot
Gyrfalcon
Lanner falcon
Peregrine falcon
Lesser kestrel
Red−legged seriema
Downy woodpecker
Carmine bee−eater
Ground hornbill
Common buzzard
White−tailed eagle
Harpy eagle
Golden eagle
Bearded vulture
King vulture
Plumbeous ibis
Eurasian spoonbill
Great cormorant
Northern gannet
Maguari stork
Humboldt penguin
Cory's shearwater
Red−throated loon
Red−billed tropicbird
Hoatzin
Herring gull
Lesser black−backed gull
Black−legged kittiwake
Common tern
Eurasian curlew
European golden plover
Northern lapwing
Takahe
Common moorhen
Common crane
Turtle dove
Collared dove
Rock dove
Band−tailed pigeon
Nicobar pigeon
Yellow−throated sandgrouse
American flamingo
MacQueen's bustard
Common cuckoo
Whiskered treeswift
Common swift
Long−billed hermit
Tawny frogmouth
European nightjar
Tufted duck
Greater scaup
Common pochard
Red−crested pochard
Mallard
Northern pintail
Common goldeneye
Long−tailed duck
Tundra swan
Mute swan
Greylag goose
Canada goose
Rock ptarmigan
Western capercaillie
Chicken
Crested tinamou
Darwin's rhea
Emu
Brown kiwi

−0.5 0.0 1.0 2.0
Technical innovation rate

B

DAR enrichment
Wilcoxon, BH FDR, 23 cell types

Negative Rho

OL ***
AST ***
MGL
VASC
OPC ***
EXC_GLU−8 ***
IOL *
INH_VIP *
EXC_GLU−5 *
EXC_GLU−3
EXC_GLU−7
INH_1
INH_SST
EXC_GLU−6
EXC_GLU−1
EXC_GLU−2
INH_LAMP5
INH_2
EXC_GLU−4
INH_STR
D2MSN
D1MSN
INH_PVALB

Positive Rho

OL
AST
MGL
VASC
OPC
EXC_GLU−8
IOL
INH_VIP
EXC_GLU−5
EXC_GLU−3
EXC_GLU−7
INH_1
INH_SST
EXC_GLU−6
EXC_GLU−1
EXC_GLU−2
INH_LAMP5
INH_2
EXC_GLU−4
INH_STR
D2MSN
D1MSN
INH_PVALB

−0.06 −0.03 0.00 0.03
Effect size

FDR < 0.05 n.s.

C

LOC107055098 (−48 bp)
ρ = 0.39, *P* = 0.011

TFAP2B (+249 bp)
ρ = 0.3, *P* = 0.004

XCL1 (−277 bp)
ρ = 0.38, *P* = 0.002

FGF10 (−353 bp)
ρ = 0.33, *P* = 0.007

TSKU (+658 bp)
ρ = 0.39, *P* = 0.003

PCK1 (−276 bp)
ρ = −0.2, *P* = 0.008

LGI1 (−338 bp)
ρ = −0.22, *P* = 0.033

GYPC (+590 bp)
ρ = −0.22, *P* = 0.004

GJC3 (−690 bp)
ρ = −0.25, *P* = 0.003

MIR1705 (+960 bp)
ρ = −0.33, *P* = 6.8×10^{-6}

Technical innovation rate

Relative evolutionary rate (RER)